

Share Base Stability and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Share Base Stability (SBS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on SBS achieves an annualized gross (net) Sharpe ratio of 0.53 (0.47), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 23 (22) bps/month with a t-statistic of 2.90 (2.89), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Growth in book equity, Net Payout Yield, Share issuance (5 year), Change in equity to assets, Momentum and LT Reversal) is 15 bps/month with a t-statistic of 2.00.

1 Introduction

Financial markets efficiently incorporate information about firms’ fundamental risks into asset prices, yet the mechanisms through which corporate actions affect investors’ consumption risk exposure remain incompletely understood. We examine how Share Base Stability (SBS), a measure capturing the consistency of firms’ equity capitalization over time, predicts cross-sectional variation in expected returns through its connection to systematic consumption risk. The stability of a firm’s share base reflects management’s financing decisions and signals the firm’s exposure to aggregate economic shocks that affect investors’ marginal utility of consumption. Despite extensive research on corporate payout policy and capital structure decisions (Fama and French, 2001; Pontiff and Woodgate, 2008), the literature lacks a comprehensive understanding of how share base stability relates to consumption-based asset pricing models and the intertemporal marginal rate of substitution.

Share base stability connects to consumption risk through multiple economic channels grounded in consumption-based asset pricing theory. Firms with unstable share bases—those frequently issuing or repurchasing equity—exhibit heightened sensitivity to aggregate consumption shocks because their financing activities correlate with macroeconomic conditions that affect investors’ marginal utility (Campbell and Cochrane, 1999; Bansal, 2004). During economic downturns when consumption growth declines and marginal utility rises, firms with volatile share bases face greater financing frictions and reduced access to capital markets, amplifying their exposure to systematic consumption risk. This mechanism aligns with long-run risk models where firms’ financing decisions reflect their exposure to persistent shocks in aggregate consumption growth (Bansal and Yaron, 2004).

The state-dependent nature of share base stability manifests through asymmetric responses to good versus bad economic times. Firms maintaining stable share bases provide insurance against consumption disasters by preserving financial flexibility

when investors' marginal utility peaks (Barro, 2006; Gabaix, 2012). These firms exhibit lower covariance with the stochastic discount factor because their operational and financial decisions remain relatively insulated from consumption growth fluctuations. Conversely, firms with unstable share bases amplify consumption risk during economic contractions, as their equity issuance and repurchase activities procyclically correlate with aggregate consumption dynamics.

The intertemporal marginal rate of substitution framework further elucidates why share base stability predicts returns. Investors demand higher compensation for holding firms whose share base volatility increases their exposure to systematic consumption risk factors (Cochrane, 2005; Lettau and Ludvigson, 2001). Share base instability signals greater sensitivity to habit formation effects, where consumption relative to past levels determines risk premia. Firms with stable share bases offer hedging benefits against adverse consumption shocks, justifying lower required returns consistent with consumption-based asset pricing models that incorporate time-varying risk aversion (Campbell and Cochrane, 1999).

Our empirical analysis reveals that a value-weighted long/short trading strategy based on Share Base Stability achieves an annualized gross Sharpe ratio of 0.53, with a monthly average excess return of 33 basis points and a t-statistic of 4.17. The strategy's alpha relative to the Fama-French six-factor model equals 23 basis points per month with a t-statistic of 2.90, demonstrating robust performance after controlling for established risk factors. The economic magnitude remains substantial across alternative portfolio construction methodologies, with the strategy earning significant alphas ranging from 18 to 35 basis points per month depending on the factor model specification.

The SBS strategy's performance persists after accounting for transaction costs, generating a net Sharpe ratio of 0.47 and maintaining statistical significance across all factor model specifications. Among the largest stocks—those exceeding the 80th

NYSE percentile—the strategy delivers an average return of 23 basis points per month with a t-statistic of 2.55, confirming that the anomaly extends beyond small, illiquid securities. The strategy’s net generalized alphas remain positive and statistically significant for all five standard factor models, indicating that SBS expands the mean-variance efficient frontier even after incorporating trading costs.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the SBS trading strategy maintains an alpha of 15 basis points per month with a t-statistic of 2.00. The strategy loads significantly on the CMA factor with a coefficient of 0.30 and t-statistic of 5.48, consistent with share base stability capturing firms’ investment patterns and their exposure to systematic consumption risk. These results position SBS in the top decile of documented anomalies, with its gross Sharpe ratio exceeding 92% of strategies in the factor zoo.

This paper advances the consumption-based asset pricing literature by establishing share base stability as a novel predictor of cross-sectional returns that captures firms’ exposure to systematic consumption risk. Unlike [Daniel and Titman \(2006\)](#) who focus on investor overconfidence in interpreting corporate actions, we demonstrate that share base stability reflects rational compensation for bearing consumption risk during adverse economic states. Our findings extend [Pontiff and Woodgate \(2008\)](#) by showing that the predictive power of equity issuance activities stems from their connection to the intertemporal marginal rate of substitution rather than purely behavioral mechanisms. While [Fama and French \(2008\)](#) document the relationship between net stock issues and returns, we provide a consumption-based theoretical foundation that explains why share base stability commands a risk premium.

Methodologically, we contribute by implementing comprehensive robustness tests following [Novy-Marx and Velikov \(2023\)](#) that account for multiple sorting procedures, transaction costs, and the proliferation of factors in the cross-section. Our analysis reveals that SBS maintains explanatory power even after controlling for 154

documented anomalies, with combination strategies incorporating SBS generating superior risk-adjusted returns. The strategy’s performance among large-cap stocks and its robustness to alternative portfolio construction methods distinguish it from anomalies that derive their strength from small, illiquid securities where implementation remains challenging.

Our results carry broader implications for understanding how corporate financing decisions interact with aggregate consumption dynamics to determine equilibrium asset prices. The documented relationship between share base stability and expected returns supports consumption-based models that emphasize state-dependent risk premia and time-varying risk aversion [Campbell and Cochrane \(1999\)](#). By linking micro-level corporate actions to macro-level consumption risk, this research bridges the gap between corporate finance and asset pricing theory, offering new insights into the economic mechanisms that drive cross-sectional return variation in rational, frictionless markets.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of share base changes and to align with the typical frequency of equity issuance and repurchase decisions. Share Base Stability signal requires two key accounting variables from COMPUSTAT. Common stock (CSTK) represents the par or stated value of common shares outstanding, reflecting the nominal value of equity shares issued by the firm. Common equity (CEQ) measures total common shareholders’ equity, which equals the sum of common stock, capital sur-

plus, retained earnings, and treasury stock adjustments, representing the book value of equity attributable to common shareholders. Both variables are fundamental components of the balance sheet and provide essential information about a firm’s equity structure. We construct the Share Base Stability signal by calculating the year-over-year change in common stock scaled by lagged common equity: $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{CEQ}(t-1)$. This ratio captures the relative change in the par value of outstanding shares, normalized by the prior year’s book equity base. An increase in this measure indicates net equity issuance activity, while a decrease suggests share repurchases or retirements. The scaling by lagged common equity ensures comparability across firms of different sizes and provides an economically meaningful measure of the magnitude of share base changes relative to the firm’s equity capital. We compute this signal using fiscal year-end values and require non-missing values for both CSTK and CEQ in consecutive years. To mitigate the influence of extreme outliers, we winsorize the signal at the 1st and 99th percentiles.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the SBS signal. Panel A plots the time-series of the mean, median, and interquartile range for SBS. On average, the cross-sectional mean (median) SBS is -0.01 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input SBS data. The signal’s interquartile range spans -0.01 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the SBS signal for the CRSP universe. On average, the SBS signal is available for 6.39% of CRSP names, which on average make up 7.69% of total market capitalization.

4 Does SBS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on SBS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high SBS portfolio and sells the low SBS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short SBS strategy earns an average return of 0.33% per month with a t-statistic of 4.17. The annualized Sharpe ratio of the strategy is 0.53. The alphas range from 0.23% to 0.35% per month and have t-statistics exceeding 2.90 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.30, with a t-statistic of 5.48 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 590 stocks and an average market capitalization of at least \$1,490 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 24 bps/month with a t-statistics of 3.24. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for eighteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 21-29bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.75. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the SBS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the SBS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and SBS, as well as average returns and alphas for long/short trading SBS strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the SBS strategy achieves an average return of 23 bps/month with a t-statistic of 2.55. Among these large cap stocks, the alphas for the SBS strategy relative to the five most common factor models range from 17 to 24 bps/month with t-statistics between 1.85 and 2.59.

5 How does SBS perform relative to the zoo?

Figure 2 puts the performance of SBS in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the SBS strategy falls in the distribution. The SBS strategy’s gross (net) Sharpe ratio of 0.53 (0.47) is greater than 92% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the SBS strategy (red line).² Ignoring trading costs, a \$1 invested in the SBS strategy would have yielded \$8.58 which ranks the SBS strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the SBS strategy would have yielded \$6.24 which ranks the SBS strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the SBS relative to those. Panel A shows that the SBS strategy gross alphas fall between the 66 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The SBS strategy has a positive net generalized alpha for five out of the five factor models. In these cases SBS ranks between the 82 and 89 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does SBS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of SBS with 210 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price SBS or at least to weaken the power SBS has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of SBS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{SBS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SBS}SBS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{SBS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on SBS. Stocks are finally grouped into five SBS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted SBS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on SBS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the SBS signal in these Fama-MacBeth regressions exceed 0.85, with the minimum t-statistic occurring when controlling for Momentum and LT Reversal. Controlling for all six closely related anomalies, the t-statistic on SBS is -1.43.

Similarly, Table 5 reports results from spanning tests that regress returns to the SBS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the SBS strategy earns alphas that range from 16-26bps/month. The

minimum t-statistic on these alphas controlling for one anomaly at a time is 2.02, which is achieved when controlling for Momentum and LT Reversal. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the SBS trading strategy achieves an alpha of 15bps/month with a t-statistic of 2.00.

7 Does SBS add relative to the whole zoo?

Finally, we can ask how much adding SBS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the SBS signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes SBS grows to \$2810.42.

8 Conclusion

This study provides compelling evidence that Share Base Stability (SBS) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Ve-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which SBS is available.

likov (2023), demonstrates that SBS-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on SBS delivers an annualized Sharpe ratio of 0.53 gross (0.47 net of transaction costs), indicating strong economic significance that survives practical implementation considerations. Most notably, the strategy generates statistically significant abnormal returns of 23 basis points per month relative to the Fama-French five-factor model plus momentum, with a t-statistic of 2.90. Even after controlling for six closely related strategies from the factor zoo—including various share issuance measures, growth metrics, and payout yields—the SBS signal maintains a significant alpha of 15 basis points per month (t-statistic = 2.00), suggesting that it captures unique information about future returns not subsumed by existing factors.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, SBS emerges as a distinct pricing anomaly that challenges traditional asset pricing models and warrants inclusion in factor model specifications. For practitioners, the signal’s robustness to transaction costs and its incremental predictive power over related strategies suggest potential value in portfolio construction and risk management applications. The relatively modest degradation from gross to net returns (approximately 6 basis points in Sharpe ratio) indicates that the strategy remains viable after accounting for real-world trading frictions.

Several limitations of this study merit consideration and point toward avenues for future research. First, while our analysis demonstrates statistical and economic significance in U.S. equity markets, the international robustness of the SBS signal remains unexplored. Future research should examine whether similar patterns exist in global equity markets and investigate potential cross-country variations in the sig-

nal’s effectiveness. Second, the economic mechanisms underlying the SBS anomaly warrant deeper investigation. While share base stability likely reflects managerial signaling, capital structure decisions, or investor sentiment, a comprehensive theoretical framework linking these mechanisms to expected returns would enhance our understanding of the phenomenon.

Additional research opportunities include examining the time-varying nature of the SBS premium, particularly during different market regimes and economic cycles. The interaction between SBS and corporate events such as mergers, acquisitions, and restructurings could provide insights into when the signal is most informative. Furthermore, investigating the signal’s performance across different firm characteristics—such as size, age, and industry—could reveal important heterogeneities in its predictive power.

In conclusion, Share Base Stability emerges as a robust and economically meaningful predictor of stock returns that adds value beyond existing factors in the literature. While further research is needed to fully understand its economic drivers and boundary conditions, the evidence presented here establishes SBS as a valuable addition to the toolkit of both researchers studying asset pricing anomalies and practitioners seeking to enhance portfolio performance.

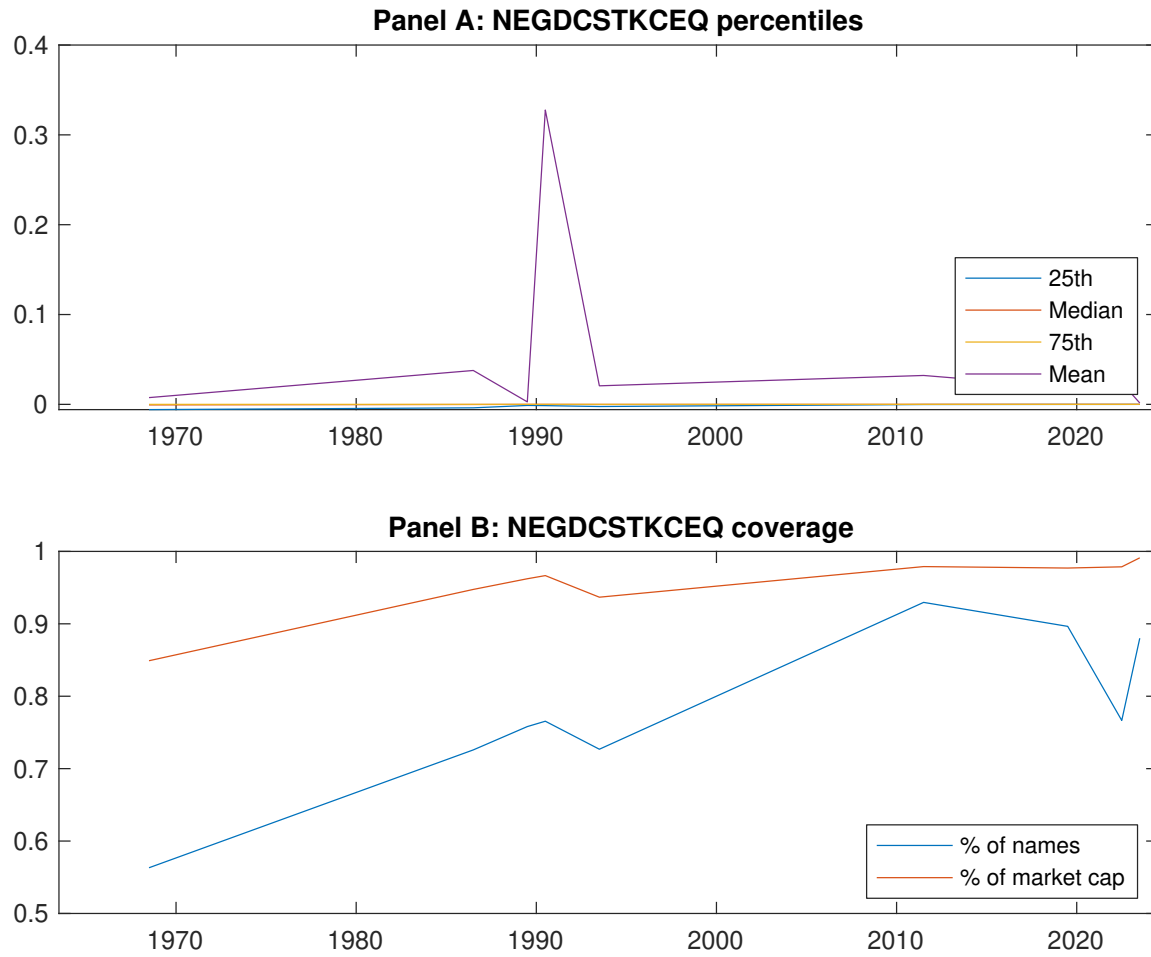


Figure 1: Times series of SBS percentiles and coverage.
This figure plots descriptive statistics for SBS. Panel A shows cross-sectional percentiles of SBS over the sample. Panel B plots the monthly coverage of SBS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on SBS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on SBS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.45 [2.70]	0.49 [2.70]	0.64 [3.63]	0.64 [3.96]	0.78 [4.85]	0.33 [4.17]
α_{CAPM}	-0.12 [-2.20]	-0.14 [-3.08]	0.03 [0.74]	0.09 [1.87]	0.23 [4.94]	0.35 [4.41]
α_{FF3}	-0.13 [-2.39]	-0.12 [-2.80]	0.05 [1.19]	0.05 [1.06]	0.19 [4.17]	0.31 [3.98]
α_{FF4}	-0.10 [-1.92]	-0.09 [-2.03]	0.05 [1.18]	0.02 [0.52]	0.17 [3.73]	0.27 [3.42]
α_{FF5}	-0.15 [-2.81]	-0.07 [-1.60]	0.08 [1.68]	-0.04 [-0.91]	0.11 [2.49]	0.26 [3.27]
α_{FF6}	-0.13 [-2.41]	-0.05 [-1.06]	0.08 [1.65]	-0.05 [-1.16]	0.10 [2.31]	0.23 [2.90]
Panel B: Fama and French (2018) 6-factor model loadings for SBS-sorted portfolios						
β_{MKT}	0.97 [76.32]	1.03 [96.13]	1.02 [91.60]	1.00 [94.31]	0.98 [90.38]	0.01 [0.49]
β_{SMB}	-0.02 [-0.90]	0.02 [1.26]	0.04 [2.45]	-0.06 [-4.21]	0.00 [0.22]	0.02 [0.72]
β_{HML}	0.07 [3.05]	-0.02 [-0.89]	-0.07 [-3.34]	0.05 [2.50]	0.04 [2.12]	-0.03 [-0.83]
β_{RMW}	0.13 [5.29]	-0.09 [-4.29]	-0.07 [-3.30]	0.12 [5.99]	0.11 [5.41]	-0.02 [-0.45]
β_{CMA}	-0.11 [-3.04]	-0.08 [-2.80]	0.01 [0.30]	0.21 [7.11]	0.19 [6.15]	0.30 [5.48]
β_{UMD}	-0.03 [-2.22]	-0.03 [-3.17]	0.00 [0.05]	0.02 [1.59]	0.01 [0.89]	0.04 [1.97]
Panel C: Average number of firms (n) and market capitalization (me)						
n	721	693	590	683	747	
me (\$10 ⁶)	1764	1490	2140	2265	2475	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the SBS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.33 [4.17]	0.35 [4.41]	0.31 [3.98]	0.27 [3.42]	0.26 [3.27]	0.23 [2.90]
Quintile	NYSE	EW	0.47 [8.45]	0.52 [9.47]	0.46 [8.78]	0.41 [7.76]	0.37 [7.19]	0.34 [6.48]
Quintile	Name	VW	0.32 [4.07]	0.34 [4.30]	0.30 [3.86]	0.27 [3.38]	0.25 [3.21]	0.23 [2.91]
Quintile	Cap	VW	0.24 [3.24]	0.26 [3.39]	0.24 [3.11]	0.20 [2.52]	0.21 [2.80]	0.18 [2.37]
Decile	NYSE	VW	0.33 [3.86]	0.33 [3.78]	0.28 [3.27]	0.24 [2.78]	0.27 [3.14]	0.24 [2.77]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.29 [3.68]	0.31 [4.00]	0.28 [3.57]	0.25 [3.27]	0.24 [3.10]	0.22 [2.89]
Quintile	NYSE	EW	0.27 [4.39]	0.33 [5.34]	0.27 [4.62]	0.25 [4.19]	0.18 [3.17]	0.17 [2.92]
Quintile	Name	VW	0.28 [3.58]	0.30 [3.84]	0.26 [3.41]	0.24 [3.15]	0.23 [2.98]	0.22 [2.81]
Quintile	Cap	VW	0.21 [2.75]	0.22 [2.90]	0.20 [2.60]	0.17 [2.29]	0.19 [2.47]	0.17 [2.25]
Decile	NYSE	VW	0.29 [3.35]	0.28 [3.28]	0.24 [2.82]	0.22 [2.57]	0.24 [2.77]	0.22 [2.58]

Table 3: Conditional sort on size and SBS

This table presents results for conditional double sorts on size and SBS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on SBS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high SBS and short stocks with low SBS. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	SBS Quintiles					SBS Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.47 [1.91]	0.68 [2.65]	0.87 [3.40]	0.90 [3.85]	0.96 [4.11]	0.52 [7.23]	0.54 [7.54]	0.51 [7.14]	0.47 [6.48]	0.44 [6.11]	0.41 [5.65]
	(2)	0.62 [2.72]	0.68 [2.87]	0.85 [3.62]	0.96 [4.26]	0.91 [4.22]	0.30 [3.55]	0.34 [4.01]	0.28 [3.39]	0.26 [3.05]	0.24 [2.88]	0.23 [2.67]
	(3)	0.55 [2.74]	0.66 [3.05]	0.74 [3.39]	0.83 [4.09]	0.95 [4.84]	0.40 [5.34]	0.41 [5.40]	0.38 [5.01]	0.37 [4.70]	0.36 [4.58]	0.35 [4.38]
	(4)	0.51 [2.74]	0.65 [3.26]	0.76 [3.77]	0.77 [4.13]	0.79 [4.32]	0.28 [3.81]	0.29 [3.99]	0.23 [3.28]	0.22 [3.12]	0.12 [1.72]	0.13 [1.78]
	(5)	0.48 [2.93]	0.47 [2.62]	0.52 [2.99]	0.51 [3.09]	0.71 [4.52]	0.23 [2.55]	0.24 [2.59]	0.22 [2.33]	0.17 [1.85]	0.21 [2.24]	0.18 [1.88]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	SBS Quintiles					SBS Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	383	384	384	382	374	31	34	40	29	29	
	(2)	109	107	108	108	107	57	56	58	57	55	
	(3)	78	78	78	77	78	99	96	98	100	100	
	(4)	66	65	66	66	66	206	204	215	213	217	
(5)	60	60	60	60	60	1479	1493	1661	1628	1887		

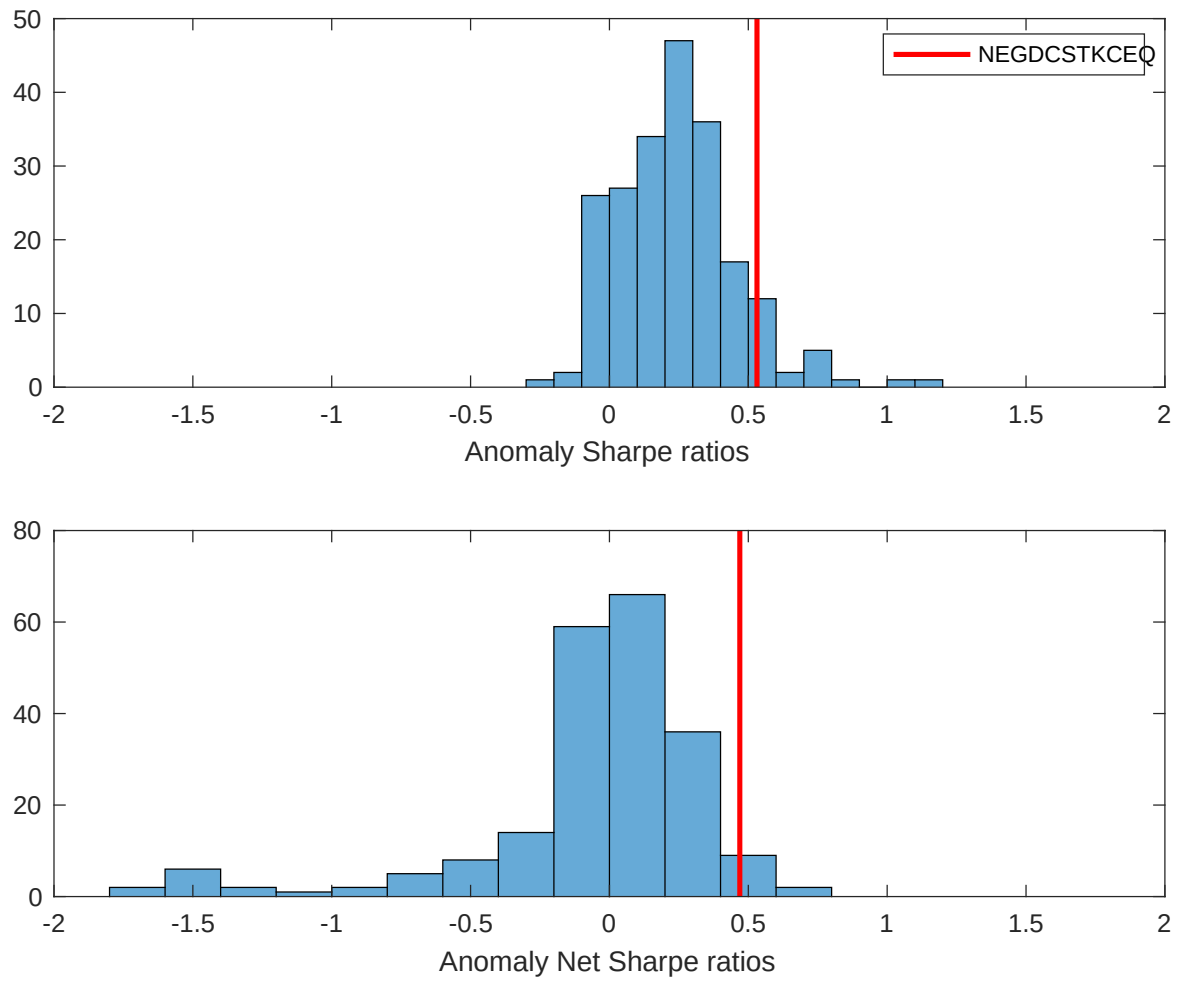


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the SBS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

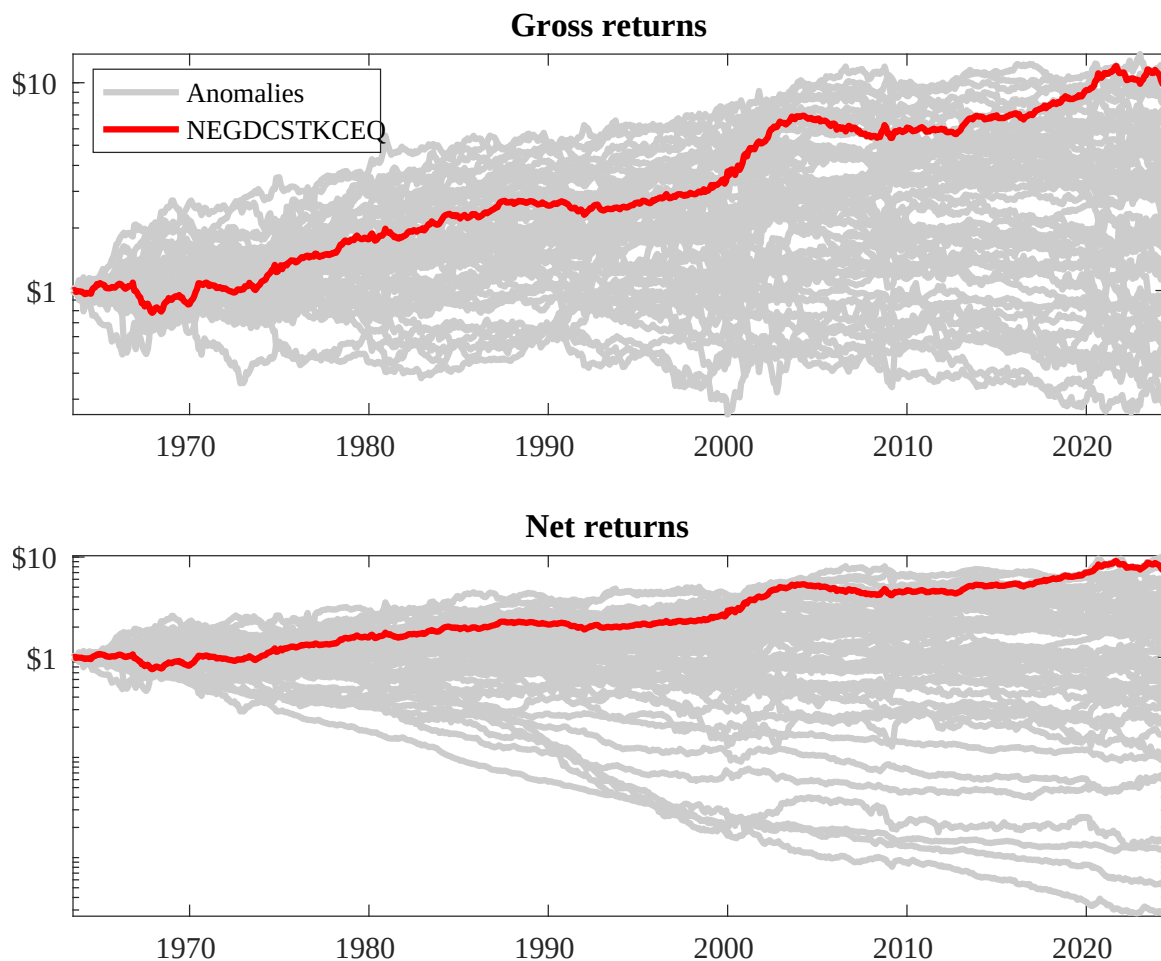
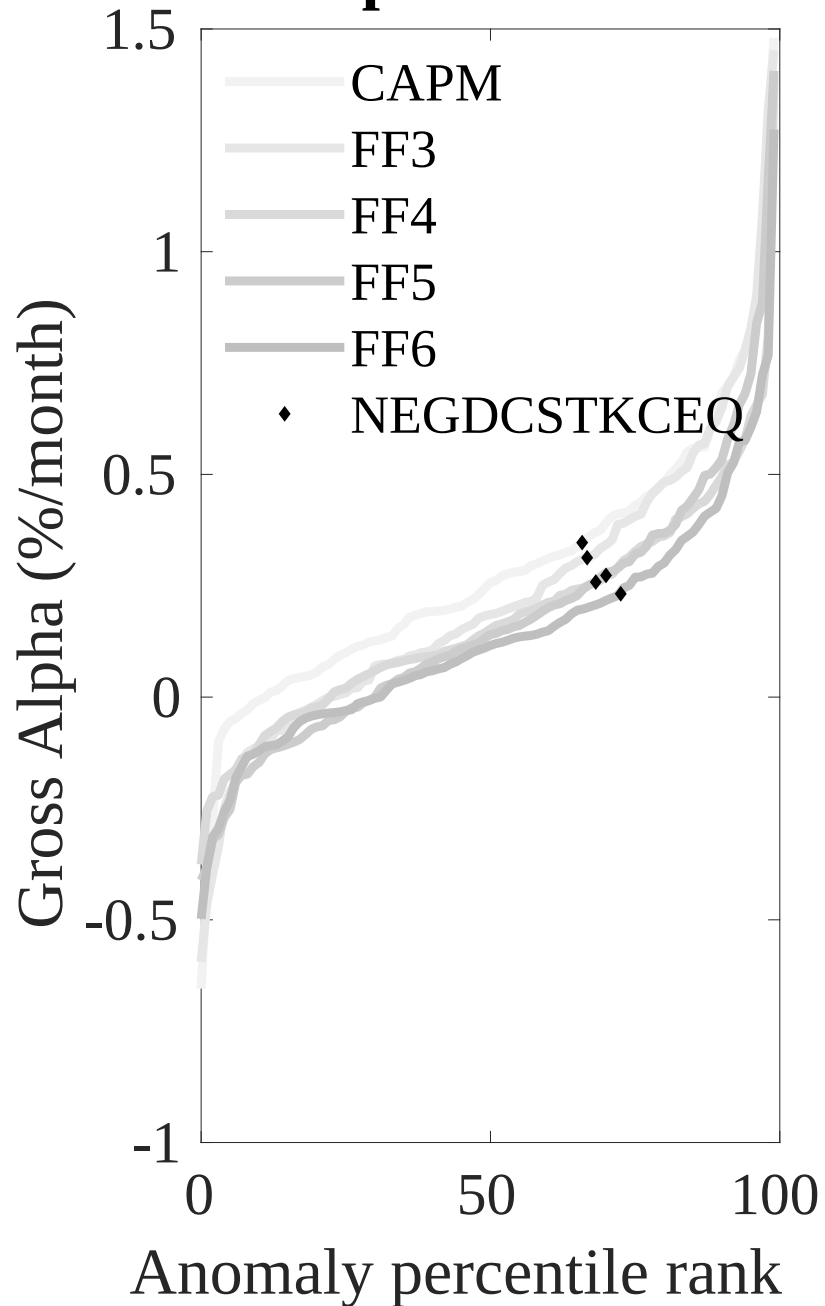


Figure 3: Dollar invested.

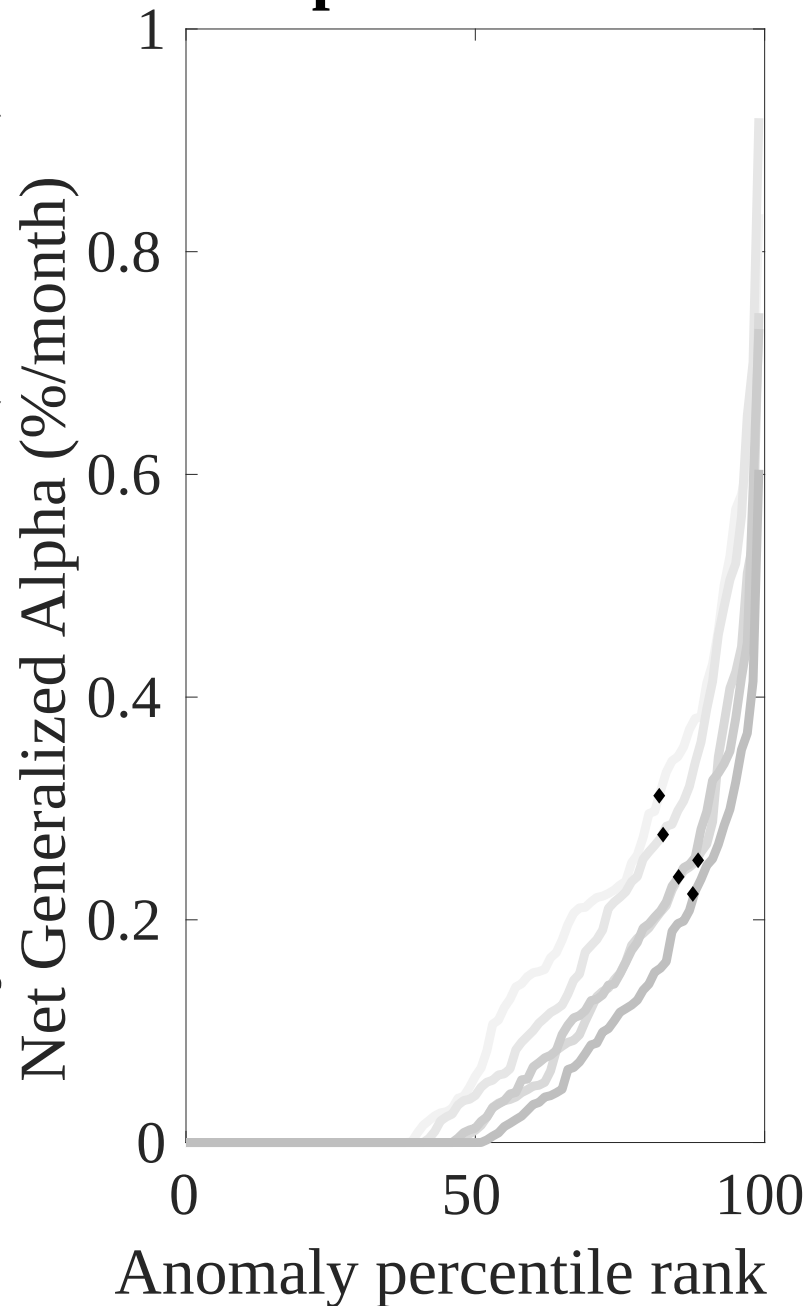
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the SBS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



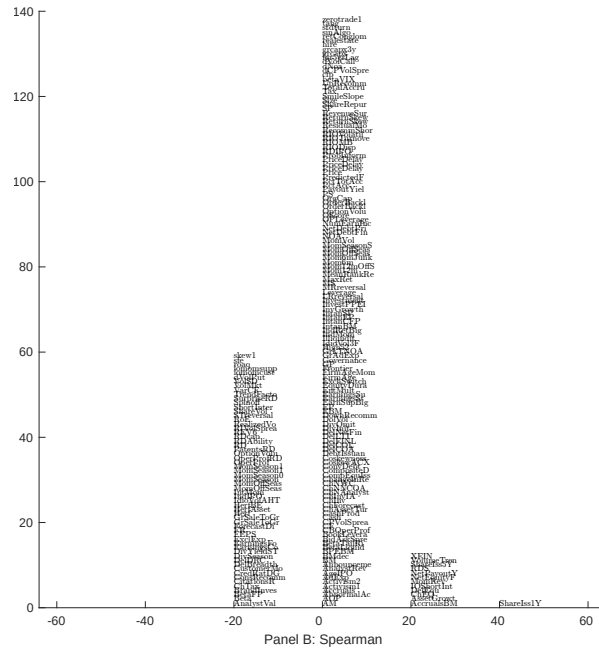
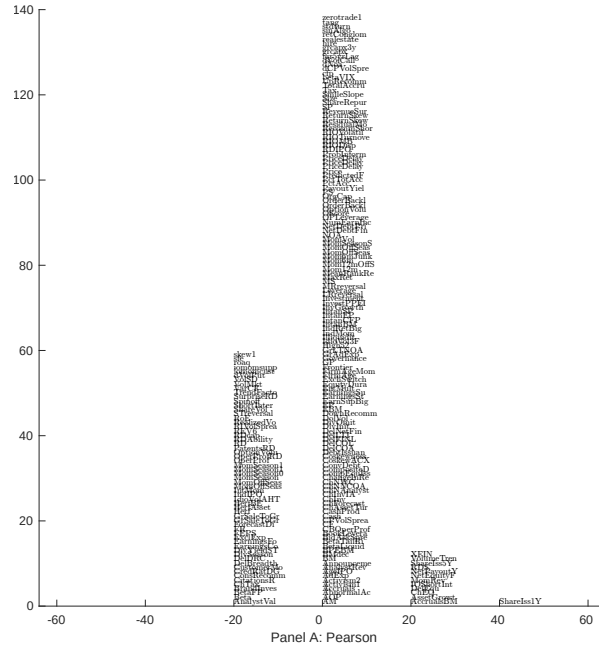


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with SBS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

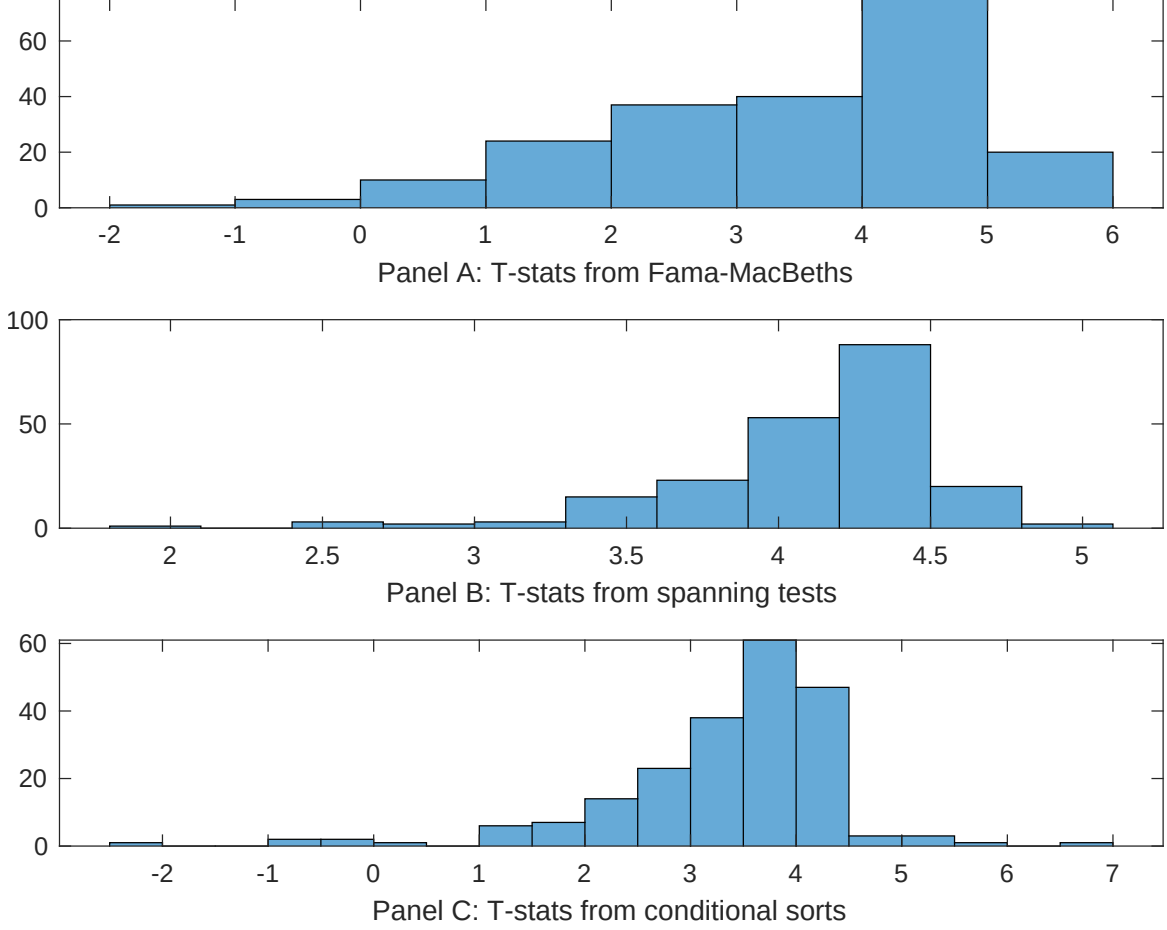


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of SBS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{SBS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SBS}SBS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{SBS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on SBS. Stocks are finally grouped into five SBS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted SBS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on SBS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{SBS}SBS_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Growth in book equity, Net Payout Yield, Share issuance (5 year), Change in equity to assets, Momentum and LT Reversal. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.04]	0.16 [6.83]	0.12 [5.43]	0.13 [6.43]	0.12 [5.68]	0.38 [1.12]	0.12 [2.73]
SBS	0.19 [3.45]	0.22 [3.64]	0.19 [2.68]	0.17 [3.07]	0.22 [3.72]	0.29 [0.85]	-0.42 [-1.43]
Anomaly 1	0.13 [5.75]						0.61 [1.21]
Anomaly 2		0.38 [3.18]					0.44 [1.83]
Anomaly 3			0.20 [1.80]				0.13 [1.48]
Anomaly 4				0.26 [4.71]			0.26 [2.32]
Anomaly 5					0.11 [3.08]		-0.13 [-1.56]
Anomaly 6						0.11 [4.27]	0.10 [3.48]
# months	715	720	715	715	720	648	510
$\bar{R}^2(\%)$	0	0	1	0	0	2	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the SBS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{SBS} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Growth in book equity, Net Payout Yield, Share issuance (5 year), Change in equity to assets, Momentum and LT Reversal. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.21 [2.79]	0.22 [2.86]	0.26 [3.32]	0.16 [2.02]	0.23 [2.96]	0.21 [2.65]	0.15 [2.00]
Anomaly 1	30.47 [7.88]						14.56 [3.14]
Anomaly 2		31.31 [7.56]					31.55 [5.39]
Anomaly 3			15.84 [5.24]				5.20 [1.60]
Anomaly 4				30.15 [7.35]			15.58 [3.38]
Anomaly 5					11.63 [2.97]		-17.59 [-3.28]
Anomaly 6						5.09 [4.77]	4.19 [4.14]
mkt	1.86 [1.02]	2.14 [1.17]	3.58 [1.87]	2.47 [1.34]	0.90 [0.48]	1.28 [0.68]	4.37 [2.41]
smb	4.54 [1.72]	2.25 [0.85]	5.88 [2.14]	5.35 [2.01]	3.15 [1.16]	1.32 [0.49]	3.60 [1.38]
hml	-2.40 [-0.68]	-5.40 [-1.52]	-7.69 [-2.02]	0.83 [0.23]	-3.07 [-0.84]	-2.81 [-0.77]	-5.82 [-1.61]
rmw	-18.17 [-4.40]	-0.17 [-0.05]	-10.87 [-2.69]	-12.47 [-3.23]	-0.58 [-0.16]	0.96 [0.26]	-16.23 [-3.76]
cma	9.97 [1.76]	-1.52 [-0.23]	15.49 [2.65]	14.26 [2.57]	17.10 [2.57]	26.36 [4.96]	-6.64 [-1.01]
umd	2.59 [1.43]	3.42 [1.89]	5.24 [2.82]	2.01 [1.10]	4.27 [2.29]	-0.97 [-0.46]	-2.39 [-1.18]
# months	728	732	728	728	732	728	728
$\bar{R}^2(\%)$	14	13	10	13	8	9	22

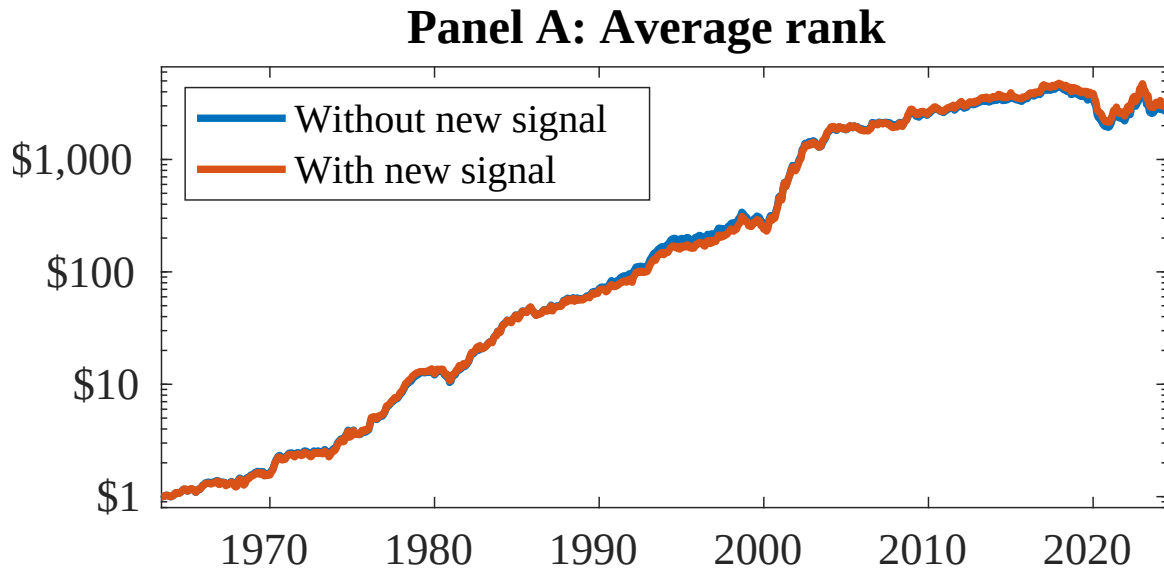


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as SBS. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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